Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 11

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Specialization	Cyber Security
Batch No	0201

Activity 1

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano menu_act1.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x menu_act1.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./menu_act1.sh
```

```
#!/bin/bash

choice=0

while [ "$choice" -ne 5 ]
do
    echo "----------------------------"
    echo "        MAIN MENU"
    echo "----------------------------"
    echo "[1] Show Today's date/time"
    echo "[2] Show files in current directory"
    echo "[3] Show calendar"
    echo "[4] Start editor to write letters"
    echo "[5] Exit/Stop"
    echo "----------------------------"
    echo -n "Enter your choice: "
    read choice

    case $choice in
        1)
            echo "Today's date and time:"
            date
            ;;
        2)
            echo "Files in current directory:"
            ls
            ;;
        3)
```

```
        3)
            echo "Calendar:"
            cal
            ;;
        4)
            echo "Opening nano editor..."
            nano letter.txt
            ;;
        5)
            echo "Exiting program..."
            ;;
        *)
            echo "Invalid choice. Please enter 1 to 5."
            ;;
    esac

    echo
done
```

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano menu_act1.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x menu_act1.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./menu_act1.sh
-----------------------------
        MAIN MENU
-----------------------------
[1] Show Today's date/time
[2] Show files in current directory
[3] Show calendar
[4] Start editor to write letters
[5] Exit/Stop
-----------------------------
Enter your choice: 1
Today's date and time:
Fri Apr 24 09:16:21 AM +0530 2026

-----------------------------
        MAIN MENU
-----------------------------
[1] Show Today's date/time
[2] Show files in current directory
[3] Show calendar
[4] Start editor to write letters
[5] Exit/Stop
-----------------------------
Enter your choice: 3
Calendar:
     April 2026
Su Mo Tu We Th Fr Sa
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
```

Activity 2

Pattern 1

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
  GNU nano 7.2                    pattern_act2.sh
#!/bin/bash

for (( i=1; i<=5; i++ ))
do
    for (( j=1; j<=i; j++ ))
    do
        echo -n "$i"
    done
    echo
done
```

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano pattern_act2.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x pattern_act2.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./pattern_act2.sh
1
22
333
4444
55555
```

Pattern 2

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
  GNU nano 7.2                    pattern2.sh *
#!/bin/bash

for (( i=1; i<=5; i++ ))
do
    for (( j=1; j<=i; j++ ))
    do
        echo -n "$j"
    done
    echo
done
```

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano pattern2.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x pattern2.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./pattern2.sh
1
12
123
1234
12345
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$
```

Pattern 3

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
  GNU nano 7.2                          pattern3.sh *
#!/bin/bash

for (( i=1; i<=5; i++ ))
do
    for (( j=1; j<=i; j++ ))
    do
        if [ "$j" -eq "$i" ]
        then
            echo -n "|_"
        else
            echo -n "| "
        fi
    done
    echo
done
```

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano pattern3.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x pattern3.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./pattern3.sh
|_
| |_
| | |_
| | | |_
| | | | |_
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$
```

Pattern 4

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
  GNU nano 7.2                                  pattern4.sh *
#!/bin/bash

for (( i=1; i<=5; i++ ))
do
    for (( j=1; j<=i; j++ ))
    do
        echo -n "* "
    done
    echo
done

for (( i=5; i>=1; i-- ))
do
    for (( j=1; j<=i; j++ ))
    do
        echo -n "* "
    done
    echo
done
```

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano pattern4.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x pattern4.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./pattern4.sh
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * *
* * *
* *
*
```

Activity 3

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
  GNU nano 7.2                          bigNum.sh *
#!/bin/bash

if [ "$#" -ne 3 ]
then
    echo "Error: Please supply exactly three numbers."
    echo "Usage: ./biggest.sh number1 number2 number3"
    exit 1
fi

num1=$1
num2=$2
num3=$3

if [ "$num1" -ge "$num2" ] && [ "$num1" -ge "$num3" ]
then
    echo "Biggest number is: $num1"
elif [ "$num2" -ge "$num1" ] && [ "$num2" -ge "$num3" ]
then
    echo "Biggest number is: $num2"
else
    echo "Biggest number is: $num3"
fi
```

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano bigNum.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x bigNum.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./bigNum.sh 10 25 15
Biggest number is: 25
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./bigNum.sh 10 25
Error: Please supply exactly three numbers.
Usage: ./biggest.sh number1 number2 number3
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ echo $?
1
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$
```

Activity 4

```
isuru@IT24100139: ~/Desktop/lab/OSSA/lab11
  GNU nano 7.2                    info_function.sh *
#!/bin/bash

showInfo()
{
    echo "Day: $(date +%A)"
    echo "Time: $(date +%T)"
    echo "Login/User: $(whoami)"
    echo "Current Directory: $(pwd)"
}

showInfo
```

```
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ nano info_function.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ chmod +x info_function.sh
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$ ./info_function.sh
Day: Friday
Time: 09:27:44
Login/User: isuru
Current Directory: /home/isuru/Desktop/lab/OSSA/lab11
isuru@IT24100139:~/Desktop/lab/OSSA/lab11$
```